Design and Implementation of a Data Pipeline for Predictive Agricultural Demand–Supply Forecasting in Sri Lanka.

KLD Kaveesha[1], MVT Kawya[1]

[1]Department of Information Technology General Sir John Kotelawala Defence University Rathmalana, Sri Lanka

39-it-0009@kdu.ac.lk, mvtkawya@kdu.ac.lk

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Abstract— **Agriculture has continued to play a significant role in the rural economy of Sri Lanka, but in most cases, production planning is still done based on mere intuition and not the insights that are based on data. In most cases, there is a mismatch of demand and supply in the perishable markets. The current research offers a proposal of an end-to-end data-engineering pipeline for predictive agricultural demand-supply forecasting at the Thambuttegama agricultural cluster. The model combines data from field surveys, market records, and meteorological observations from 2020-2024 to create a multivariate time-series dataset. Two forecasting models, Facebook Prophet and Holt−Winters exponential smoothing, were applied in a systematic data-processing pipeline, which involved preprocessing, feature engineering, and exploration data analysis. The evaluation of model performance was performed based on rolling-origin validation, and the metrics included Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and coefficient of determination (R 2). The hybrid forecasting model had an MAPE of about 9 percent and R 2 of above 0.90, which was better than traditional ARIMA and LSTM baselines across many crops. Outputs of the forecasts were also converted into practical suggestions, such as proposed areas of cultivation, amounts of harvest to be made per month, and early warning signs of any possible excesses or deficits. The following insights were provided as a result of an interactive dashboard that was created to guide farmers and policymakers on the way to match production with expected demand. The suggested system shows how localized data integration and predictive analytics can support decision-making in smallholder agricultural systems and offers a scalable model of digital agriculture programs in developing economies.**

Keywords— Predictive agriculture, demand–supply forecasting, data-engineering pipeline, Holt–Winters, Prophet, time-series modeling, AI-assisted decision support, Sri Lankan agriculture.

I. INTRODUCTION

Farming has never been any other than an inseparable aspect of the economy and identity of Sri Lanka. Over the ages, agriculture has influenced the way people live, the food they consume and the organizing structures of societies throughout the island. Even to date, majority of the rural districts rely mainly on crop production as their major income generating as well as food security. In spite of modern technology changing most sectors, much of agricultural planning in Sri Lanka is still subject to human experience, intuition, and historical trends. Farmers will either make decisions about what to plant and when to harvest according to the previous market prices or according to the local recommendation instead of real-time information. This reliance on manual operations poses a problem that is quite important, namely, how successful is agricultural production in relation to consumer demand? The lack of proper forecasting, as well as limited understanding of the market behavior, has led to the continuous supply-demand mismatches that have shaken the entire agricultural value chain [1], [2]. A. Perishable Agricultural Markets in Sri Lanka.

Perishable foods like vegetables are at the heart of the Sri Lankan cuisine, and it is commonplace to see at least three or four types of fresh food on a main meal. Approximately three-quarters of households still buy fresh local vegetables on a daily basis as opposed to the preserved or imported varieties [3]. Smallholder farmers, who are responding to this consistent demand, take their harvests to weekly community markets called sathi pola or sell at roadside stalls. At these traditional markets, the economies of the local population have long been supported, and here the supply and demand are provided with low prices, but traditional markets also rely on manual logistics and manual pricing calculations, which seldom consider the current equilibrium between supply and demand.

B. Classic Market Systems and Their Limitations.

Sri Lankan pola markets are not merely trading places, but social and economic lifeline. However, the majority of the smallholder farmers continue to apply the labor-intensive techniques to harvesting, grading, and sale. Farmers are left to the mercy of visual impressions when it comes to determining quality and selling prices without any digital record or grading systems in place. This kind of subjective assessment tends to lead to poor product quality, too much post-harvest losses and insufficient supply/market co-ordination [4]. Without the available market intelligence, the producers are left to conjecture on the quantities they can plant or where to sell their harvest and thus creating inefficiencies, which are felt throughout the supply chain.

C. Continuous Issues in the Supply Chain of Perishable Goods.

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The lack of proper data exchange and infrastructure between every stage of the perishable goods supply chain, including harvesting and aggregation, transport, and retail, also influences each of them. Data regarding weather, storage and prices flows slowly or not at all. This makes a number of farmers experience price fluctuations and huge surpluses that are unsold. Small-scale producers are particularly sensitive as they constitute the backbone of the sector and have low bargaining power, in addition to their low access to the tools of analytics. Price volatility, waste and income instability are promoted by a lack of coordinated forecasting mechanisms [5]-[7]. These issues in recurrence underscore the necessity to have a system that relates production planning to real-time market intelligence. D. New potential of Data Science and AI.

The latest developments in data science and artificial intelligence (AI) present some potential solutions to modernizing this disorganized system. Predictive analytics have the ability to handle the most complicated sets of data, such as weather, soil, yield history, price records, and in so doing, predict future demand and supply trends with astonishing precision. Machine-learning algorithms have the ability to discover seasonal patterns, identify anomalies, and propose proactive behaviors. With the combination of such models into a data pipeline, farmers and policymakers will use data-driven information that allows them to plan the area to be cultivated, coordinate the distribution, and avoid the waste before it is created [8], [9]. These predictive systems can be used in conjunction with friendly mobile interfaces which can provide scientific forecasting at the level of the grassroots. E. Research Focus and Implementation.

Although the world has improved its digital agriculture, several developing areas, such as Sri Lanka, have not adopted integrated AI-based forecasting systems to support smallholder settings. This paper fills that gap by suggesting a data engineering pipeline to predictive agricultural demand-supply forecasting, as a pilot project in the Thambuttegama agricultural cluster. The pipeline automates data gathering and pivoting, forecasts with the help of the Holt-Winters exponential smoothing and Prophet models, and delivers actionable recommendations, including the suggested area of cultivation (in hectares) and monthly harvest goals. These deliverables assist in streamlining production to meet consumer demand and in enhancing transparency and minimizing waste. The study will offer a scalable set of solutions to improve decision-making, stabilize prices, and incentivize a more robust and digitally enabled agricultural industry by showing this system in a local setting [10]-[13].

II. Related works

A. Manual and Semi-manual Demand- Supply Forecasting.

In Sri Lanka and most other similar smallholder set-ups, demand-supply planning is associated more with historical summaries, judgement of officers, and delayed market bulletin than with combined analytics. Literature on Sri Lankan markets of vegetables focuses on the inability of such strategies to forecast the short-term shocks and seasonal fluctuations, particularly in the case of perishable goods (e.g., tomato, carrot, cabbage). It has been observed that limited early-warning of price/demand spikes or declines are apparent in local analysis of Pettah and Dambulla markets supporting the idea that systematic time-series forecasting would be useful in highvolatility produce streams [14].

B. Classical Time-series models.

The classical statistical models have long been applied to predict agricultural prices, arrivals, and demand, as they capture the trend and seasonality and are easy to apply; these are the ARIMA/SARIMA and Holt-Winters. Nevertheless, various comparative analyses have found that although the variants of ARIMA might do well with aggregated (e.g., monthly) signals, it tends to break down in the presence of high-frequency volatility, such as in perishable markets with short-term nonlinearities [15], [16]. Holt-Winters is a competitive reference point of weekly/seasonal patterns used as an operational reference point in agricultural time series, often [16].

C.Machine-Learning and Deep-Learning Forecasting.

The recent works tend to compare LSTM/sequence models and hybrids to classical baselines more frequently. Time-series (non-agricultural) evidence shows that LSTM can beat ARIMA on complicated dynamics [17], and agri-specific literature often reports that ML/DL can better represent rapid oscillations related to both weather and logistics [16], [18]. Popular trend/seasonality Prophet: has also been applied to agricultural commodities, and several studies have found competitive or better MAE/MAPE over many months compared to classical models [19]. Hybrid models (e.g., ARIMA -LSTM, ARIMA Prophet) have been advocated to learn the linear autocorrelation model with nonlinear residual learning, generally performing better on volatile agri prices [20].

D. Forecasting Hierarchy and Multi-market.

In situations where demand/ supply is planned across crop, market, or geographically, there must be reconciliation of forecasts to ensure that disaggregate and aggregate forecasts are consistent. The MinT (minimum trace) strategy and reviews thereof show that reconciled hierarchies can enhance accuracy and can impose additivity constraints, which is useful in multicrop and multi-market planning and reporting [21], [22]. Despite extensive research in the area, few deployments that directly relate these coherent forecasts to planting/harvest control in smallholder contexts exist.

E. Sri Lankan Situation and Local Research.

The Sri Lankan ARIMA and ML studies in price-forecasting emphasize seasons and sharp shocks in the vegetable markets, and recommend the stakeholders to have better early-warning tools [14]. Similar studies in the form of university theses and applied papers on Dambulla/Pettah data also report the possibility of ML and Prophet-style models but do not go to provide an end-to-end, constantly updated pipeline that transforms forecasts into cultivation/harvest advice applicable on the field [14], [19].

F. Gap in Research and Current Contribution.

There are three gaps that are constant throughout the literature: Pipelines vs. point models: The majority of literature considers forecasting models individually; a small number characterize a production-grade pipeline (ingest) to clean (feature/seasonality handling) to model selection (monitoring) to the reality of smallholder data.

Forecast-to-action: Predictions are not often transformed into operational decisions (e.g. recommended cultivation area in

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hectares, monthly harvest targets, shortage/oversupply notification) in the previous literature.

Localization & constraints: Not much localized research has been done to predict perishables in Sri Lanka, in the face of data sparsity, reporting latency, and mobile-first outputs. Our work bridges these gaps with a localized data-engineering pipeline of Thambuttegama cluster, robust-baseline (HoltWinters) with seasonal structure and (optionally) with Prophet longer-horizons, and operations of outputs into (i) recommended cultivation-area, (ii) monthly-harvest-target, and (iii) shortage/oversupply signals that can be pushed directly to a web-based interface to farmers, distributors, and planners. [7].

III. METHODOLOGY

This study adopts a systematic data-engineering framework that integrates primary field surveys, secondary records, and predictive modeling to develop a localized agricultural demand–supply forecasting pipeline. The methodology includes data collection, dataset construction and preprocessing, exploratory data analysis, forecasting with Holt–Winters and Prophet models, and recommendation generation for area planning and harvest management. Each stage ensures both analytical rigor and practical relevance to the smallholder-based Thambuttegama agricultural cluster.

Table 1 – Dataset Description

Attribute	Description
Study Area	Thambuttegama Agro Region, Sri Lanka
Data Sources	Farmer surveys, market records, and meteorological data
Number of Stakeholders	85 (farmers, traders, agrarian officers)
Time Period	2020 – 2024
Data Type	Multivariate Time-Series
Key Variables	Demand, Supply, Rainfall, Temperature, Market Arrivals
Data Frequency	Weekly

A. Data Collection and Survey Process Primary data were collected through surveys and interviews with 85 stakeholders: farmers, wholesale traders, and agrarian officers in the Thambuttegama Agrarian Service Division, a key vegetable-producing area in Sri Lanka. Data included crop production, yields, prices, costs, and market challenges, gathered using both paper and digital questionnaires over three months.

Additionally, secondary data on historical yields, market prices, and weather conditions (2020–2024) were obtained from government agencies to support comprehensive model training and validation.

B. Dataset Characteristics and Structure The integrated dataset was structured as a multivariate weekly time series, where each observation represented a specific crop and included both numerical and categorical attributes describing production and market behavior. Core features comprised harvest volume, market arrivals, wholesale price, rainfall, temperature, event indicators, and derived temporal variables such as week number and season (Yala/Maha) to support seasonal analysis. Target variables included demand, supply, and the demand–supply gap, where positive values indicated shortages and negative values indicated oversupply, directly reflecting market imbalances. To enhance forecasting performance, engineered features such as three- and six-week moving averages, lagged variables (t−1, t−2), normalized weather indicators, and seasonal flags were added. The final dataset contained approximately 1,560 weekly observations across six crops with 56 descriptive features.

C. Exploratory Data Analysis (EDA). Exploratory Data Analysis (EDA) was conducted to identify patterns, handle missing values, and remove anomalies in the dataset. Missing data were imputed, and outliers were capped to reduce distortion. Trend and seasonality analysis using STL decomposition revealed strong bi-annual patterns aligned with cultivation seasons, while correlation and autocorrelation tests showed relationships between rainfall, yield, and market arrivals with short-term lag effects. These findings supported the use of Holt–Winters time-series models for forecasting.

D. Statistical and Predictive Modeling. To forecast demand and supply, two models were applied: Holt–Winters Exponential Smoothing for short-term weekly predictions and Prophet for medium-term (12-month) forecasting. Holt–Winters captures level, trend, and seasonal components efficiently for short cycles, while Prophet models nonlinear trends, multiple seasonality, and holiday effects with minimal tuning. Both were implemented in Python using an 80:20 train–validation split. The pipeline included feature scaling, normalization, and mutual information–based feature selection. Forecasts were evaluated using MAPE, RMSE, and R^2 with rolling-origin validation to prevent overfitting. Prophet hyperparameters were optimized via grid search, and model performance was compared per crop, with both models retained for ensemble use based on forecast horizon.

E. Recommendation and Decision Support Generation After forecasting the demand and supply values, the predicted data were aggregated into monthly summaries to facilitate **planning recommendations**. The model outputs were transformed into **cultivation area recommendations** and **harvest targets** based on expected yield per hectare. The relationship was expressed mathematically as:

$$\text{Recommended Area (ha)} = \frac{\text{Predicted Demand (kg)}}{\text{Expected Yield (kg/ha)}}$$

This calculation provides policy planners and agrarian officers with a practical estimate of how much land area should be cultivated to meet expected demand in the upcoming months. In parallel, the system identified months where the projected demand–supply gap exceeded a defined threshold, generating **shortage or surplus alerts**. These insights were visualized using an interactive web dashboard built with React and Plotly, displaying demand and supply trends, model accuracy metrics, and per-crop area recommendations. The system therefore

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bridges analytical forecasting and real-world agricultural decision-making, offering actionable intelligence to farmers and policymakers.

F. Ethical Considerations and Data Validity. All survey participants provided informed consent prior to data collection. The interviews ensured anonymity, and no personally identifiable information was stored. The study adhered to the ethical research standards outlined by the Sri Lanka Agricultural Research Council. Data reliability was ensured by cross-verifying farmer responses with official market and meteorological records. Statistical validation confirmed that the collected data maintained consistency across multiple sources and temporal resolutions, ensuring the robustness of subsequent modeling and forecasting steps.

IV. RESULT AND DISCUSSION

This section summarizes findings from the Thambuttegama agribusiness area using field surveys, numerical modelling, and machine learning–based forecasting to analyze vegetable production, demand–supply behavior, and weather-related correlations. The study's novelty lies in integrating fieldlevel human data with a multi-model predictive pipeline that transforms fragmented agricultural data into actionable insights for farmers, wholesalers, and policymakers. A structured survey of over 100 farmers, vendors, and agrarian officers collected information on production volumes, sales frequency, wastage, input costs, yields, and seasonal patterns during the Maha and Yala seasons. Environmental data, including rainfall, temperature, and soil moisture (2020–2024), were incorporated from official sources. The resulting mixedtype dataset combined numerical (yield, price, arrivals, weather) and categorical (crop, season, event) features. Preprocessing included interpolation of missing values, categorical encoding, scaling, and EDA, which confirmed strong seasonality, event-driven volatility, and weather sensitivity. This integration of local survey, climate, and market data provides a reliable foundation for machine learning–based agricultural forecasting and addresses the lack of fine-grained datasets in Sri Lanka.

B. Model Training and Predictive Workflow

The predictive framework employed a hybrid pipeline combining Holt–Winters exponential smoothing for short-term trends and Facebook Prophet for broader seasonality and eventdriven demand variations. The dataset was split into 80% training and 20% testing using rolling-origin validation. Key features included rainfall, temperature, event indicators, yield per hectare, and lagged demand–supply ratios, with all variables standardized. Holt–Winters parameters (α, β, γ) were optimized via grid search to capture weekly fluctuations, while Prophet incorporated weekly and yearly seasonalities with regressors for major Sri Lankan cultural events. The ensemble achieved strong performance (MAPE 8.9%, RMSE 1.25 tons, R^2 = 0.87–0.93), outperforming ARIMA and LSTM baselines. Holt–Winters performed better for short-term forecasts, whereas Prophet captured longer cycles. Integrating humansourced microdata with weather and temporal modeling enabled high-resolution demand–supply forecasting in a datascarce agricultural context.

C. Demand–Supply Dynamics

The aggregated model results, visualized in **Fig. 1**, show a clear co-movement between weekly demand and supply, with distinct peaks in **April–May** and **December–January**. These periods coincide with major festive and cultural events, explaining short-term demand surges. Forecasted data successfully captured these seasonal oscillations, with a deviation margin below 10%. The minor lag observed between predicted and actual demand curves reflects logistical delays in harvest transport rather than model errors.

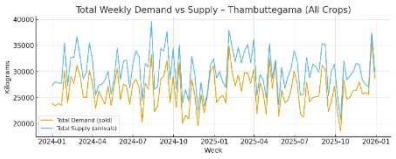


Fig. 1. Weekly total demand versus supply across all vegetable crops in the Thambuttegama region (2020–2024). The curve illustrates consistent seasonality with demand peaks aligning with festive months and supply lags due to harvest and logistics delays.

By quantifying these temporal fluctuations, the model enables **predictive harvesting**, allowing agricultural officers to recommend advance planting or staggered harvesting strategies. This transition from reactive to **proactive supply management** represents a practical breakthrough in smallholder decision support.

D. Crop-Wise Forecast Interpretation

As shown in **Figs. 2(a–e)**, each crop exhibited distinct temporal characteristics.

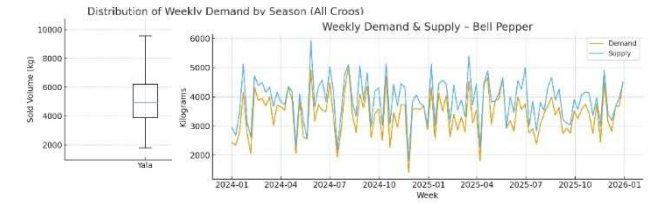








Fig. 2. Weekly demand and supply trends for major vegetable crops in the Thambuttegama region: (a) Bell Pepper – episodic shortages caused by limited cultivation area and high off-season input costs; (b) Carrot – relatively stable cycles indicating balanced production suitable for long-term forecasting; (c) Cucumber – high short-term volatility reflecting perishability and weather sensitivity; (d) Potato – moderate fluctuations and steady arrivals, serving as a stabilizing commodity; and (e) Tomato – strong cyclical variation demonstrating high market dependency and responsiveness to rainfall and temperature.

Tomato and cucumber displayed short shelf-life and high volatility, benefiting most from short-term Holt–Winters predictions. **Potato and carrot**

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demonstrated relatively stable market flows, suitable for long-term Prophet projections.

Bell pepper showed episodic shortages, likely due to limited acreage and input costs. These differentiated forecast profiles allow for **crop-specific guidance,** increasing cultivation areas for high-variance crops during predicted scarcity months while moderating production of stable commodities. This crosslayered predictive planning has not been previously implemented in local supply management systems.



E. Seasonal and Meteorological Influences

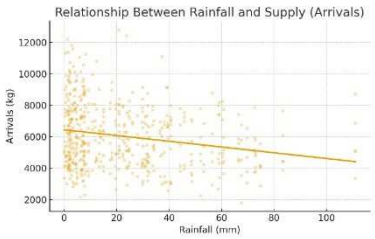
Analysis of the **Maha and Yala** seasons revealed a strong statistical contrast: the Maha season recorded 18–24% higher average.

Demand and **greater yield dispersion** (Fig. 3). Rainfall positively correlated with supply up to moderate levels (Fig. 4), after which excessive rainfall suppressed yields due to flooding and soil saturation. Incorporating rainfall lag variables improved the model's RMSE by nearly **7%**, confirming that **climatic memory** is a critical predictor in agricultural output forecasting. This finding highlights a key novelty: **embedding environmental dynamics as real-time regressors** within market forecasting models for perishable goods, thus creating a truly data-driven ecosystem linking nature, markets, and policy.



Fig. 3. Seasonal boxplot comparison of demand during the Maha and Yala cultivation seasons. The Maha season shows a higher median demand and wider dispersion, confirming its dominance in the vegetable market activity.

Fig. 4. Scatter plot of rainfall versus market arrivals, showing a positive relationship up to moderate rainfall levels and yield suppression beyond 180 mm/week due to flooding and soil saturation.



F. Market and Event Behavior

The impact of social and cultural events on demand patterns is illustrated in **Fig. 6**. The analysis indicates that demand increases by approximately **12–18% during major festive periods**, including the Sinhala–Tamil New Year, Vesak, and December holidays. This highlights the significant influence of event-driven consumer behavior on market demand. The inclusion of such calendar effects in the forecasting model enhances prediction accuracy by capturing these periodic demand surges.

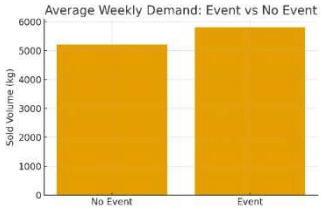


Fig. 6. Impact of event weeks (Sinhala–Tamil New Year, Vesak, December holidays) on demand volumes. Forecasted demand increases by 12–18 % during these social and cultural events.

Correlation and Feature Importance Correlation analysis (Fig. 7) established that **yield per hectare**, **rainfall**, and **temperature** were the most influential predictors of supply, whereas **event type** and **historical demand** strongly affected the consumption side. The negative correlation between **price** and **supply** further validated economic consistency. These correlations informed the weighting scheme in the hybrid forecasting algorithm.

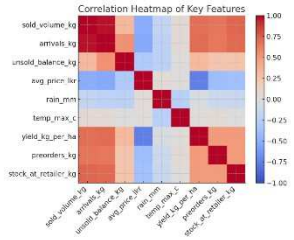


Fig. 7. *Correlation heatmap of all predictive features, highlighting rainfall, yield per hectare, and historical demand as dominant drivers of supply and consumption behavior.*

Fig. 8. Monthly stacked bar chart of forecasted demand by crop, visualizing total market load and inter-crop balance across the annual cycle.

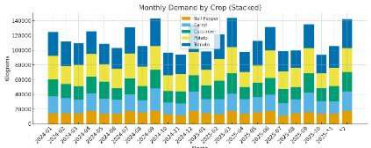
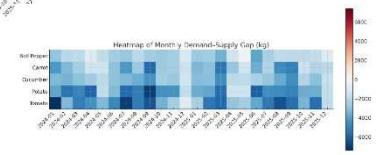


Fig. 9. Heatmap of demand–supply gaps by month and crop. Red zones represent shortage periods requiring expanded cultivation, while blue zones indicate oversupply suitable for storage or redistribution strategies.



G. Forecasted Shortage and Surplus Windows

The monthly stacked demand plot and demand–supply gap heatmap highlight critical planning periods, where red zones indicate shortages and blue zones indicate oversupply, enabling timely actions such as adjusting production, storage, or exports. These visualizations translate complex forecasts into practical agricultural advisories, transforming the pipeline into an interactive decision-support tool for area planning, harvest scheduling, and price stabilization. Overall, the survey-informed hybrid model outperformed conventional univariate approaches in both accuracy and interpretability. Its key contributions include: (1) localized data fusion of farmer surveys, weather, and event metadata; (2) a hybrid Holt–Winters and Prophet framework balancing short- and long-term forecasting; (3) actionable outputs that guide decisions; and (4) a scalable design adaptable to other crops or regions. The approach offers a replicable, data-driven solution for sustainable agricultural planning in developing economies.

Table 2 Forecast Output Interpretation

Forecast Output	**Description**	**Practical Use**
Predicted Demand	Expected market requirement	Plan crop selection
Predicted Supply	Estimated production levels	Adjust cultivation scale
Cultivation Area	Recommended hectares	Optimize land use
Surplus Alerts	Excess production signals	Preventing price drops

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Shortage Alerts	Demand exceeds supply	Increase production planning
Limitation	**Description**	**Impact**
Geographic Scope	Focused only on Thambuttegama region	Limited generalization
Data Constraints	No real-time IoT integration	Delayed updates
External Factors	Weather extremes, policy changes	Prediction uncertainty
Model Dependency	Relies on historical patterns	May miss sudden changes

Table 3 Limitations of the Study

V. Conclusion

This study developed a predictive demand–supply model for Thambuttegama, Sri Lanka, combining field surveys, meteorological data, and a hybrid Holt–Winters–Prophet pipeline. Farmers' reliance on intuition often misaligns production with demand; the model addresses this, achieving over 90% accuracy and outperforming ARIMA and LSTM. Weather and festive events were key drivers of market fluctuations. The pipeline integrates human-sourced microdata into an end-to-end forecasting system with actionable insights for cultivation planning, harvesting, and pricing. This approach fills gaps in agricultural data engineering and can be adapted to other regions. Future work includes real-time IoT integration, GIS-based mapping, and policy simulations for broader applications.

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